

Prasanna B

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EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, Tamil Nadu

B.Tech in Computer Science and Engineering — CGPA: 8.70 | 4th Sem SGPA: 8.42

Aug 2024 – May 2028

- Coursework: Data Structures, Algorithms, Operating Systems, DBMS, Linear Algebra, COA, Full Stack Web Development

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, TypeScript, Haskell **Backend:** FastAPI, REST APIs, SQLAlchemy

Frontend: React, Vite, HTML5, CSS3, Tailwind CSS, GSAP, Framer Motion, Lenis, Lucide React

Databases: SQLite, Neo4j, Supabase, Cloudinary **Tools:** Git, Linux, Bash, Docker

Core Concepts: DSA, Graph Algorithms, Probabilistic Data Structures, System Design

EXPERIENCE

Web Developer Intern | CODEFLUXZ

May 2026 – Present

AI Company Website Development *Remote*

- Designed and developed codefluxz.com using React, Vite, and Tailwind CSS with a dark-themed UI; deployed on Vercel with GoDaddy DNS.
- Implemented GSAP ScrollTrigger animations, Framer Motion transitions, 3D laptop hero scroll animation with CSS transforms, and Lenis smooth scroll.

NSS Unit-3 Student Coordinator

Aug 2024 – Present

Amrita Vishwa Vidyapeetham

Coimbatore, Tamil Nadu

- Led community outreach initiatives, sustainability programs, and environmental welfare events across campus and local communities.
- Coordinated volunteer teams for rural development activities and campus-level awareness campaigns.

PROJECTS

dupegun | Python, SHA-256, Bloom Filters, TUI, REST

2026 

- Cross-platform CLI tool for duplicate file detection using a 3-pass hashing engine (size grouping, partial hash, full SHA-256); published on PyPI.
- Features interactive TUI, browser-based web UI, real-time folder monitoring, perceptual image dedup, plugin system, and JSON/CSV/HTML export.

Eco-Fix | React, Firebase, JavaScript, HTML, CSS

2025 

- Responsive web app for eco-friendly waste management with Firebase Authentication and Firestore for real-time data handling.
- Designed interactive React UI with modern frontend techniques to enhance accessibility and user experience.

G-Wallet | FastAPI, PostgreSQL

2026 

- Secure digital wallet with ACID-compliant transaction handling, concurrency-safe batch processing, and admin verification workflow.
- Implemented authentication, audit logging, and modular REST APIs with robust backend validation mechanisms.

FleetTrack | ESP32, React, Node.js

2026 

- Full-stack real-time fleet tracking system integrating ESP32 GPS hardware with a React live dashboard for vehicle location and telemetry.
- Implemented route visualization, backend telemetry processing, and scalable architecture for continuous fleet monitoring.

Bloom Filter URL Filtering System | Python, HTML

2025 

- Custom Bloom Filter evaluating probabilistic URL filtering performance with analysis of memory vs. false-positive trade-offs.
- Demonstrated efficiency of probabilistic data structures for scalable, high-volume lookup environments.

Mines | Java, C++, Graph Algorithms

2026 

- Algorithm-driven Minesweeper engine using deterministic solving strategies, constraint propagation, and graph traversal.
- Designed modular architecture for extensible board sizes; implemented logical inference for AI-based solution validation.

PUBLICATIONS

Towards Sustainable Charcoal Production from *Prosopis Juliflora*

Integrating Local Knowledge and Clean Technology  SSRN  Scopus

2025

- Applied PRA tools (resource mapping, seasonal calendars, stakeholder consultations) in Adancheri village, Tamil Nadu to assess environmental, health, and socioeconomic impacts of traditional earth mound kilns.
- Proposed improved kiln technologies, capacity-building initiatives, and sustainable resource management strategies for local communities.